

Overview

AI engineers frequently integrate external services and serve models through APIs. Understanding HTTP and JSON is essential for reliable model serving, microservice integration, and production monitoring.

HTTP Basics

Common HTTP methods: - **GET**: read data. - **POST**: create data or request an action. - **PUT**: replace/update data. - **DELETE**: remove data.

Important status codes: - 200 success - 400 bad request - 401 unauthorized - 404 not found - 500 server error

JSON Basics

JSON represents structured data using key-value pairs, arrays, and nested objects. It is the most common payload format in modern APIs.

REST API Patterns

REST APIs organize access around resources and endpoints, for example: - `/posts` - `/users/1`

Query parameters (e.g., `?userId=1`) help filter and paginate results.

AI Engineering Use Cases

- Calling external APIs (LLM providers, feature services, metadata services).
- Serving model predictions through REST endpoints consumed by product applications.

Common Pitfalls

- Not handling non-200 responses and timeouts correctly.
- Assuming response shape is always valid JSON without checks.

Interview Prep Checklist

- Describe a simple REST endpoint design and request/response flow.
- Explain how to call an API from Python and parse JSON safely.
- Explain how to add retries, timeout handling, and response validation.